Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 8: Approved Solutions

Disjunctive Normal Form

Covers Lehman, Leighton & Meyer, *MCS* Section 3.4.1

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first. Fill truth tables on this page.

Part I: How to Practice This Material

Count first, write second

The recipe, and the check that goes with it

One AND-term per **true** row. In each term, a variable appears *plain* if it is T in that row and *negated* if it is F.

Before writing anything, count the T's in the column. That is exactly how many AND-terms you will have. If you finish with a different number, you have made an error — and you will know immediately rather than after checking every row.

Three mistakes to expect

1. **Building from the false rows.** That produces the CNF instead. False rows are Friday.
2. **Omitting a variable from a term.** In DNF every AND-term names every variable.
3. **Expecting DNF to be short.** It usually is not. Normal form means standard shape, not small.

Part II: Practice Problems

Section A — Reading DNF off a table

1. Build the truth table for $(\sim P) \vee Q$, then write its DNF.

P	Q	$(\sim P) \vee Q$	AND-term
T	T	T	$P \wedge Q$
T	F	F	—
F	T	T	$(\sim P) \wedge Q$
F	F	T	$(\sim P) \wedge (\sim Q)$

Approved solution

$$(\sim P) \vee Q \;\equiv\; (P \wedge Q) \;\vee\; ((\sim P) \wedge Q) \;\vee\; ((\sim P) \wedge (\sim Q))$$

Three true rows, three AND-terms.
Worth noticing: this is the same DNF we built in class for $P \Rightarrow Q$ — which it must be, since Theorem 1.48 says the two formulas are equivalent. **Equivalent formulas have identical DNFs.** That is not a coincidence; it is the whole reason normal forms are useful.

2. Write the DNF of each. You may work from a table or reason directly.

(a) $P \wedge Q$

(b) $\sim Q$ (over the variables P and Q)

(c) $P \vee Q$

Approved solution

(a) $P \wedge Q$. True in one row only, so the DNF is a single AND-term — and it is the formula itself. Some formulas are already in DNF.
(b) $(P \wedge (\sim Q)) \vee ((\sim P) \wedge (\sim Q))$. True in two rows. Note that the DNF must mention P even though the original formula does not — *every* AND-term names *every* variable in the agreed list.
(c) $(P \wedge Q) \vee (P \wedge (\sim Q)) \vee ((\sim P) \wedge Q)$. Three true rows, three terms.

Part (b) is the instructive one. Students often write just $\sim Q$ and stop. That is a correct *formula* but not a DNF over $\{P, Q\}$, and the distinction matters for comparing two formulas term by term.

Section B — Three variables

3. Build the DNF of $(P \wedge Q) \vee R$.

P	Q	R	$(P \wedge Q) \vee R$	AND-term
T	T	T	T	$P \wedge Q \wedge R$
T	T	F	T	$P \wedge Q \wedge (\sim R)$
T	F	T	T	$P \wedge (\sim Q) \wedge R$
T	F	F	F	—
F	T	T	T	$(\sim P) \wedge Q \wedge R$
F	T	F	F	—
F	F	T	T	$(\sim P) \wedge (\sim Q) \wedge R$
F	F	F	F	—

Approved solution

Five true rows, so five AND-terms:

$$(P \wedge Q) \vee R \equiv (P \wedge Q \wedge R) \vee (P \wedge Q \wedge (\sim R)) \vee (P \wedge (\sim Q) \wedge R) \vee ((\sim P) \wedge Q \wedge R) \vee ((\sim P) \wedge (\sim Q) \wedge R)$$

The original is four symbols long; the DNF has fifteen literals. That is normal. A DNF is a *canonical* form, not a compressed one.

Section C — Using DNF to test equivalence

4. Write the DNF of each formula, then say whether the two are equivalent.

(i) $\sim(P \Rightarrow Q)$ (ii) $P \wedge (\sim Q)$

Approved solution

(i) $\sim(P \Rightarrow Q)$ is true only in the row P true, Q false. So its DNF is the single term $P \wedge (\sim Q)$.
(ii) $P \wedge (\sim Q)$ is already a single AND-term naming both variables, so it is its own DNF.
Same DNF, so the two are equivalent — which is Theorem 1.50 from Week 2.

The method matters more than the answer. To test whether two formulas are equivalent, put both in DNF, sort the terms, and compare. No cleverness required, and it always terminates. Contrast that with hunting for a chain of laws, which works only if you spot the right rewrite.

5. Two students propose conditions for granting access:

Ana: $(P \wedge Q) \vee (P \wedge (\sim Q))$ Ben: $P \wedge (Q \vee (\sim Q))$

Are they equivalent? Answer by comparing DNFs, and say in one sentence what both conditions really amount to.

Approved solution

Ana is already in DNF: two terms, $(P \wedge Q)$ and $(P \wedge (\sim Q))$.
Ben: $Q \vee (\sim Q)$ is a tautology, so Ben's condition is true exactly when P is. Its DNF over $\{P, Q\}$ is $(P \wedge Q) \vee (P \wedge (\sim Q))$ — the same two terms.
Equivalent. Both amount simply to P: the value of Q is irrelevant to either condition.

Neither student wrote P, and both wrote something that looks like it depends on Q. Putting both into DNF exposes that they do not. This is exactly the kind of thing a code reviewer should catch.

Section D — Writing

6. In five to eight complete sentences, explain why every compound statement has a DNF, and why two formulas are equivalent exactly when their DNFs match. Refer to the recipe.

Approved solution

Model answer. Any compound statement built from n variables has a truth table with 2^n rows, and its behaviour is completely determined by which of those rows come out true. The recipe builds one AND-term for each true row, choosing each variable plain or negated so that the term is true in that row and false in every other. ORing those terms therefore produces a formula true in exactly the chosen rows and false elsewhere, which is the same column the original formula had — so the DNF is equivalent to it, and every formula has one. For the second question, two formulas are equivalent precisely when their columns agree, and the recipe turns a column into a DNF and back without loss. So matching columns give matching DNFs, and matching DNFs give matching columns. That makes comparing sorted DNFs a complete test for equivalence, requiring no ingenuity at all.

Full credit requires the link between "the column determines the formula" and "the recipe reconstructs the column." An answer that only restates the recipe earns about half.

Section E — Challenge (optional)

7. For n variables, what is the largest possible number of AND-terms in a DNF? Which formulas achieve it, and what formula has the *fewest*?

Approved solution

The largest possible number is 2^n — one term for every row — and this happens exactly when the formula is a **tautology**, since then every row is true.
The fewest is **zero**, which happens exactly when the formula is a **contradiction**: no true rows, so no terms. We write that DNF as **F**.

The pairing to notice. On Friday you will see that the CNF is built from the *false* rows, so a tautology has an empty CNF and a contradiction has a CNF with all 2^n clauses. The two normal forms are at their largest and smallest in opposite cases.

Instructor notes

- **Problem 1** lands the key idea: $(\sim P) \vee Q$ and $P \Rightarrow Q$ have the *same* DNF because they are equivalent. Say this out loud — it is the reason normal forms exist.
- **Problem 2(b)** is the diagnostic. Students who answer "$\sim Q$" have not absorbed that a DNF over $\{P, Q\}$ mentions P in every term.
- **Problem 5** is the one to put on the board. Two plausible-looking access conditions, both of which are secretly just P. It makes the case for normal forms better than any abstract argument.
- **Problem 7** sets up Friday's contrast directly. Assign it if you want the CNF lecture to land faster.
- **Chartrand does not cover this material.** Point students to *MCS* Section 3.4.1 on Canvas; there are no textbook exercises to assign for self-check, so the self-check items are written into the Lecture 8 notes instead.